Stochastic Dynamical Models
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Contents

1 Introduction

Definition 1.1 (Probability Space). *A* ***probability space*** *is a measure space* $(\Omega, \mathscr{F}, P)$, *where the measure* P *satisfies* $P(\Omega) = 1$

- Ω*:* ***Sample space*** *(set of all possible outcomes or results of that experiment).*
- $\mathscr{F}$*:* ***Event space****. Event = subset of outcomes of an experiment to which the probability is assigned.*
- *For* $A \in \mathscr{F}$, $P(A)$*:* ***Probability function****. Assigns to each event in the event space a probability in* $[0, 1]$.

Example: Coin flip.

$$\text{Sample Space: } \{H, T\}, \quad \text{Random Variable: } X(H) = 1,\, X(T) = 0, \quad \text{Probability: } P(\{H\}) = P(\{T\}) = \frac{1}{2}.$$

$$\texttt{Sample Space} \longrightarrow \texttt{Random Variable} \longrightarrow \texttt{Probability}$$

Definition 1.2 (Sample Space Ω). Ω *is the set of all possible outcomes of the experiment.*

Definition 1.3 (Event Space $\mathscr{F}$). $\mathscr{F}$ *is a collection of subsets of* Ω, *called* ***events****.It is said that* $\mathscr{F}$ *is a* σ*-algebra on* Ω.

Requirements for an event space *(or* σ*-algebra) from* Ω*:*

(i) Contains both the empty set and Ω*:* $\emptyset, \Omega \in \mathscr{F}$.

(ii) Closed under complement: if $A \in \mathscr{F}$, *then* $A^c \in \mathscr{F}$.

(iii) Closed under (countable) union: if $A_i \in \mathscr{F}$ *for* $i \in \mathbb{N}$, *then* $\displaystyle\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$.

Conceptually, the event space is the collection of subsets of outcomes of an experiment to which probabilities are asigned.

Definition 1.4 (Probability)**.** *The probability measure P is a function $P : \mathscr{F} \to [0,1]$ that assigns to each event $\mathscr{A} \in \mathscr{F}$ a real number $P(A)$ called this a* ***probability****, satisfying;*

(a) For all $A \in \mathscr{F}$, $0 \le P(A) \le 1$.

(b) $P(\Omega) = 1$.

(c) If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$. Generally

Give a set of of disjoint events $A_i \cap A_j = \emptyset$ for $i \neq j$.

$$P\left(\bigcup_{i=1}^{\infty} E_k\right) = \sum_{i=1}^{\infty} P(E_k).$$

Definition 1.5 (Random Variable)**.** *A random variable X maps from Ω to another measurable space $(S, \mathscr{T})$:*

$$X : \Omega \to S.$$

$$X^{-1}(S') = \{\omega \in \Omega \mid X(\omega) \in S'\}.$$

$$P_X(S') = P(X^{-1}(S')) = P(\{\omega \in \Omega \mid X(\omega) \in S'\}).$$

If there is some subset S' of S whose probability we want to reason about, we measure it by looking at the preimage under X of that subset of Ω.

AÑADIR TODAS LAS IMAGENES Y DEFINICIONES ASOCIADAS A LA VARIABLE ALEATORIA

2 Discrete-Time Markov Chains

Definition 2.1 (Stochastic Process)**.** *Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let I be a (countable) state space. A* ***(discrete-time) stochastic process*** *with values in I is a family*

$$\{X_n\}_{n \in \mathbb{N}_0}, \qquad X_n : \Omega \to I.$$

For $i \in I$ we write $\mathbb{P}(X_n = i) = \mathbb{P}\{\omega \in \Omega : X_n(\omega) = i\}$.

Definition 2.2 (Markov Chain and Markov Property)**.** *A stochastic process $\{X_n\}_{n \in \mathbb{N}_0}$ with values in I is a* ***Markov chain*** *if for all $n \in \mathbb{N}_0$ and all $i_0, \ldots, i_{n-1}, i, j \in I$,*

$$\mathbb{P}(X_{n+1} = j \mid X_n = i,\ X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

We denote the ***one-step transition probabilities*** *by*

$$p_{ij}(n) := \mathbb{P}(X_{n+1} = j \mid X_n = i), \qquad i, j \in I.$$

The chain is ***time-homogeneous*** *if $p_{ij}(n)$ does not depend on n; then we write p_{ij}.*

Definition 2.3 (Transition Matrix and Distributions)**.** *In the time-homogeneous case, the matrix $P = (p_{ij})_{i,j \in I}$*

*is the **transition matrix** and satisfies*

$$p_{ij} \geq 0, \qquad \sum_{j \in I} p_{ij} = 1 \quad \textit{for every } i \in I \quad \textit{(each row is a probability distribution).}$$

*If $\mu^{(0)}$ is the **initial distribution** on I, $\mu_i^{(0)} = \mathbb{P}(X_0 = i)$, then the law at time n is*

$$\mu^{(n)} = \mu^{(0)} P^n, \qquad \mu_j^{(n)} = \mathbb{P}(X_n = j) = \sum_{i \in I} \mu_i^{(0)} (P^n)_{ij}.$$

Theorem 2.1 (Chapman–Kolmogorov)**.** *For $m, n \in \mathbb{N}$ and $i, j \in I$, let $p_{ij}^{(n)} := \mathbb{P}(X_n = j \mid X_0 = i)$. Then*

$$p_{ij}^{(m+n)} = \sum_{k \in I} p_{ik}^{(m)} p_{kj}^{(n)}, \qquad \textit{equivalently} \qquad P^{m+n} = P^m P^n.$$

Remark 1 (Meaning of a State Transition)**.** *A state $i \in I$ represents a possible configuration of the system. The entry p_{ij} is the probability that, in one time step, the system moves from state i to state j. A row of P lists all possible next states starting from a fixed current state.*

Examples

(1) Fair coin as a Markov chain. State space $I = \{H, T\}$. With independent flips,

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad \mu^{(n)} = \mu^{(0)} P^n.$$

Each row is the same because the next outcome does not depend on the current one.

(2) Two–state weather model. Let $I = \{\mathrm{S}, \mathrm{R}\}$ (sunny/rainy). For $0 \leq a, b \leq 1$,

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

Here $a = \mathbb{P}(\text{sun} \to \text{rain})$ and $b = \mathbb{P}(\text{rain} \to \text{sun})$.

(3) Simple random walk on $\mathbb{Z}$. State space $I = \mathbb{Z}$. For fixed $p \in [0, 1]$ and $q = 1 - p$,

$$p_{i,i+1} = p, \qquad p_{i,i-1} = q, \qquad p_{ij} = 0 \text{ otherwise.}$$

Thus P is tridiagonal with constant super/sub-diagonals.

(4) n-step transitions. For a time-homogeneous chain,

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i) = (P^n)_{ij}, \qquad \sum_{j \in I} p_{ij}^{(n)} = 1 \text{ for each } i \in I.$$